

Лабораторная работа 1
«Оптимизация доступа к памяти, циклов и ветвлений»

Выполнил работу:

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ИС-341

Установка пакетов

```
mrk@DESKTOP-MBCTIMN:~$ sudo apt install linux-tools-generic
[sudo] password for mrk:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libpython2-stdlib libpython2.7-minimal libpython2.7-stdlib python-pkg-resources python-setuptools python2 python2-minimal python2.7 python2.7-minimal
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  hwdmdata linux-tools-5.15.0-134 linux-tools-5.15.0-134-generic linux-tools-common
The following NEW packages will be installed:
  hwdmdata linux-tools-5.15.0-134 linux-tools-5.15.0-134-generic linux-tools-common linux-tools-generic
0 upgraded, 5 newly installed, 0 to remove and 72 not upgraded.
Need to get 8228 kB of archives.
After this operation, 29.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 hwdmdata all 0.357-1 [26.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-tools-common all 5.15.0-134.145 [268 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-tools-5.15.0-134 amd64 5.15.0-134.145 [7930 kB]
11% [3 linux-tools-5.15.0-134 914 B/7930 kB 0%]
```

```
mrk@DESKTOP-MBCTIMN:~$ sudo apt install build-essential libomp-dev
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.9ubuntu3).
The following packages were automatically installed and are no longer required:
  libpython2-stdlib libpython2.7-minimal libpython2.7-stdlib python-pkg-resources python-setuptools python2 python2-minimal python2.7 python2.7-minimal
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libllvml4 libomp-14-dev libomp5-14
Suggested packages:
  libomp-14-doc
The following NEW packages will be installed:
  libllvml4 libomp-14-dev libomp-dev libomp5-14
0 upgraded, 4 newly installed, 0 to remove and 72 not upgraded.
Need to get 24.7 MB of archives.
After this operation, 118 MB of additional disk space will be used.
```

Поиск папки установившегося пакета

apparmor	environment.d	kernel	networkd-dispatcher	rsyslog	ubuntu-release-upgr
ader	file	linux-tools	openssh	sasl2	udev
apt	gcc	linux-tools-5.15.0-134	os-release	snappd	ufw
bfd-plugins	girepository-1.0	llvm-14	pam.d	software-properties	usrmerge
binfmt.d	git-core	locale	pm-utils	ssl	valgrind
byobu	gnupg	lsb	policykit-1	sysctl.d	wsl
cnf-update-db	gnupg2	man-db	polkit-1	systemd	x86_64-linux-gnu
command-not-found	gold-ld	mime	python2.7	sysusers.d	
compat-ld	groff	modprobe.d	python3	tc	
console-setup	hdpam	modules	python3.10	terminfo	
cpp	init	modules-load.d	python3.11	tmpfiles.d	
dbus-1.0	initramfs-tools	netplan	recovery-mode	ubuntu-advantage	
dpkg					

```
mrk@DESKTOP-MBCTIMN:~/lab1/cache-dgemm$ perf stats ls
WARNING: perf not found for kernel 5.15.167.4-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-5.15.167.4-microsoft-standard-WSL2
  linux-cloud-tools-5.15.167.4-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
```

После ошибки установили пакет valgrind

Задание:

1. Определить ускорение и число промахов кеш-памяти версии с порядком циклов ikj (dgemm_interchange)

2. Определить ускорение и число промахов кеш-памяти блочной версии (dgemm_block),

определить размер блока, на котором достигается наименьшее время работы

3. Оценить ускорение блочной версии с раскрученным циклом и автоматической векторизацией (dgemm-unroll.c)

Вывод простого перемножения

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-dgemm.sh
[sudo] password for hope332:
```

```
# DGEMM def: N=1024, elapsed time (sec) 5.566810
```

```
Performance counter stats for 'taskset --cpu-list 0 ./dgemm':
```

```
<not supported>      branch-misses
```

```
16.740132178 seconds time elapsed
```

```
16.703309000 seconds user
```

```
0.019980000 seconds sys
```

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ callgrind_annotate callgrind.out.50455
```

```
-----
Profile data file 'callgrind.out.50455' (creator: callgrind-3.18.1)
-----
```

```
I1 cache:
```

```
D1 cache:
```

```
LL cache:
```

```
Timerange: Basic block 0 - 3265105205
```

```
Trigger: Program termination
```

```
Profiled target: ./dgemm (PID 50455, part 1)
```

```
Events recorded: Ir Bc Bcm Bi Bim
```

```
Events shown: Ir Bc Bcm Bi Bim
```

```
Event sort order: Ir Bc Bcm Bi Bim
```

```
Thresholds: 99 0 0 0 0
```

```
Include dirs:
```

```
User annotated:
```

```
Auto-annotation: on
```

```
-----
Ir          Bc          Bcm          Bi          Bim
-----
132,277,770,340 (100.0%) 3,247,406,964 (100.0%) 3,288,172 (100.0%) 2,097,527 (100.0%) 178 (100.0%) PROGRAM TOTALS
-----
```

- Ir — общее число выполненных инструкций.

- Bc — общее число инструкций ветвления.

- Bcm — число неправильных предсказаний ветвлений (branch misses).

Вывод ikj

```
[sudo] password for hope332:
# DGEMM interchange: N=1024, elapsed time (sec) 1.851254

Performance counter stats for 'taskset --cpu-list 0 ./dgemm':

    <not supported>          branch-misses

    5.592715902 seconds time elapsed

    5.587000000 seconds user
    0.000000000 seconds sys
```

```
I1 cache:
D1 cache:
LL cache:
Timerange: Basic block 0 - 3265105310
Trigger: Program termination
Profiled target: ./dgemm (PID 53977, part 1)
Events recorded: Ir Bc Bcm Bi Bim
Events shown:    Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds:      99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir          Bc          Bcm          Bi          Bim
-----
132,277,770.672 (100.0%) 3,247,407.050 (100.0%) 3,288,187 (100.0%) 2,097,535 (100.0%) 178 (100.0%) PROGRAM TOTALS
```

Блочное умножение

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-dgemm.sh
[sudo] password for hope332:
# DGEMM bloc: N=1024, BS=4, elapsed time (sec) 2.923994

Performance counter stats for 'taskset --cpu-list 0 ./dgemm':

    <not supported>          branch-misses

    8.810011742 seconds time elapsed

    8.803822000 seconds user
    0.000000000 seconds sys
```

```

Profile data file 'callgrind.out.55009' (creator: callgrind-3.18.1)
-----
I1 cache:
D1 cache:
LL cache:
Timerange: Basic block 0 - 6427144875
Trigger: Program termination
Profiled target: ./dgemm (PID 55009, part 1)
Events recorded: Ir Bc Bcm Bi Bim
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir Bc Bcm Bi Bim
-----
151,282,473,763 (100.0%) 5,355,433,416 (100.0%) 302,524,850 (100.0%) 2,097,535 (100.0%) 178 (100.0%) PROGRAM TOTALS

```

```

hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-dgemm.sh
# DGEMM bloc: N=1024, BS=8, elapsed time (sec) 2.371543

```

Performance counter stats for 'taskset --cpu-list 0 ./dgemm':

```

<not supported>      branch-misses

7.156135738 seconds time elapsed

7.124424000 seconds user
0.020012000 seconds sys

```

```

hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ callgrind_annotate callgrind.out.56530
-----
Profile data file 'callgrind.out.56530' (creator: callgrind-3.18.1)
-----
I1 cache:
D1 cache:
LL cache:
Timerange: Basic block 0 - 4633636380
Trigger: Program termination
Profiled target: ./dgemm (PID 56530, part 1)
Events recorded: Ir Bc Bcm Bi Bim
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir Bc Bcm Bi Bim
-----
144,197,259,403 (100.0%) 4,159,761,089 (100.0%) 459,465,249 (100.0%) 2,097,534 (100.0%) 178 (100.0%) PROGRAM TOTALS

```

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-
[sudo] password for hope332:
# DGEMM bloc: N=1024, BS=16, elapsed time (sec) 3.956636

Performance counter stats for 'taskset --cpu-list 0 ./dgemm':

<not supported>      branch-misses

11.945327279 seconds time elapsed

11.769900000 seconds user
0.060101000 seconds sys
```

```
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir              Bc              Bcm              Bi              Bim
-----
141,273,045,896 (100.0%) 3,670,526,285 (100.0%) 214,847,841 (100.0%) 2,097,534 (100.0%) 178 (100.0%) PROGRAM TOTALS
-----

Ir              Bc              Bcm              Bi              Bim      file:function
-----
141,087,561,066 (99.87%) 3,650,642,307 (99.46%) 214,708,478 (99.94%) .          .          ./dgemm.c:dgemm_block
[/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm/dgemm]
```

```
# DGEMM bloc: N=1024, BS=64, elapsed time (sec) 2.386691

Performance counter stats for 'taskset --cpu-list 0 ./dgemm':

<not supported>      branch-misses

7.205744280 seconds time elapsed

7.165854000 seconds user
0.020044000 seconds sys
```

```
LL cache:
Timerange: Basic block 0 - 3409052474
Trigger: Program termination
Profiled target: ./dgemm (PID 58997, part 1)
Events recorded: Ir Bc Bcm Bi Bim
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir              Bc              Bcm              Bi              Bim
-----
139,311,725,734 (100.0%) 3,343,371,822 (100.0%) 51,270,603 (100.0%) 2,097,534 (100.0%) 178 (100.0%) PROGRAM TOTALS
-----
```

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-dgemm.sh
# DGEEM bloc: N=1024, BS=256, elapsed time (sec) 3.410408

Performance counter stats for 'taskset --cpu-list 0 ./dgemm':

    <not supported>      branch-misses

    10.273318691 seconds time elapsed

    10.258422000 seconds user
    0.009998000 seconds sys
```

```
LE Cache:
Timerange: Basic block 0 - 3293555864
Trigger: Program termination
Profiled target: ./dgemm (PID 59789, part 1)
Events recorded: Ir Bc Bcm Bi Bim
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir          Bc          Bcm          Bi          Bim
-----
138,849,767,261 (100.0%) 3,266,374,079 (100.0%) 12,771,755 (100.0%) 2,097,534 (100.0%) 178 (100.0%) PROGRAM TOTALS
```

1. Ускорение = $5.56/1.85 = 3$ раза
число промахов кеш-памяти версии с порядком циклов `ikj`
(`dgemm_interchange`) = 3 288 187 (в простом перемножении
3 288 172)
2. Ускорение = $3.96/2.37 = 1.67$ раза
и число промахов кеш-памяти блочной версии (`dgemm_block`),=
6 554 770, наименьшее время 2.37 достигается при `BS=8`
3. Раскручивание цикла
Без раскрутки

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-dgemm-unroll.sh
1024 4.357263 0.49

Performance counter stats for 'taskset --cpu-list 0 ./dgemm-unroll':

    <not supported>      branch-misses:u

    4.371106403 seconds time elapsed

    4.349337000 seconds user
    0.009998000 seconds sys

Auto-annotation: on

-----
Ir          Bc          Bcm          Bi          Bim
-----
57,441,368,192 (100.0%) 1,686,201,794 (100.0%) 42,012,078 (100.0%) 377 (100.0%) 177 (100.0%) PROGRAM TOTALS
```

Раскрученный

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/cache-dgemm$ ./run-dgemm-unroll.sh
1024 0.092768 23.15

Performance counter stats for 'taskset --cpu-list 0 ./dgemm-unroll':

    <not supported>      branch-misses:u

    0.103817587 seconds time elapsed

    0.095236000 seconds user
    0.000000000 seconds sys

Events recorded: Ir Bc Bcm Bi Bim
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on

-----
Ir          Bc          Bcm          Bi          Bim
-----
1,468,905,421 (100.0%) 37,807,346 (100.0%) 4,230,260 (100.0%) 373 (100.0%) 179 (100.0%) PROGRAM TOTALS
```

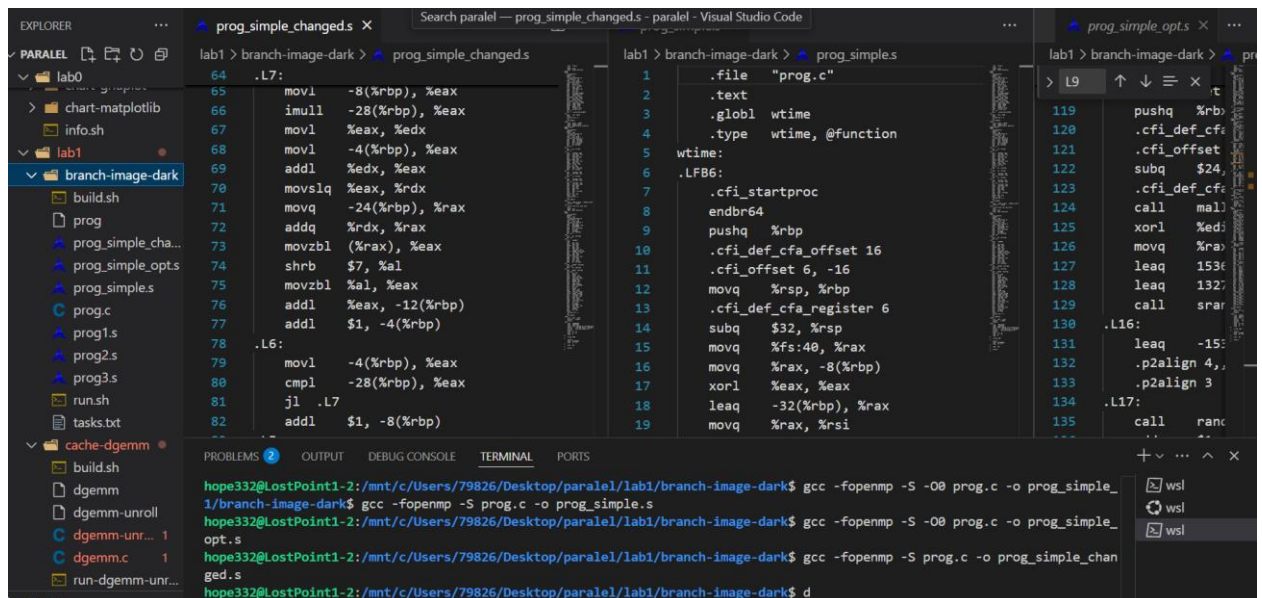
Ускорение 4.357263/0.092768= 46.88 раза

Задание:

1. Устранить ветвление в цикле функции `lower_bound_opt()`
2. Проверить устраняет ли компилятор ветвление при использовании ключа `-O2`

```
int image_is_dark_opt(uint8_t *img, int width, int height)
{
    int count = 0;
    for (int i = 0; i < height; i++)
    {
        for (int j = 0; j < width; j++)
        {
            count += (img[i * width + j] >> 7);
        }
    }
}
```

Заменили ветвление на арифметическую операцию



Видно, что кусочки кода при `O2` и замене ветвления в коде одинаковые значит произошла оптимизация, а в варианте с ветвлением происходит прыжок (75 строка)

Задание:

- * Устранить зависимость по данным в развернутом цикле (функция `vec_sum_v3()`)
- * Оценить ускорение по сравнению с функцией `vec_sum()` и `vec_sum_v2()`
- * Сохраняется ли ускорение при включении оптимизации `-O2`?

1. Устранили зависимость по данным

```
int vec_sum_v3(int *vec, int n)
{
    // TODO: remove data dependency in unrolled loop
    unsigned int i;
    int s = 0, t1 = 0, t2 = 0, t3 = 0;
    for (i = 0; i + 3 < n; i += 4)
    {
        s = s + vec[i];
        t1 = t1 + vec[i + 1];
        t2 = t2 + vec[i + 2];
        t3 = t3 + vec[i + 3];
    }
    t1 += t2 + t3;
    for (; i < n; i++)
    {
        s += vec[i];
    }
    return s + t1;
}
```

2. Оценка ускорения
`Vec_sum()`

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/pipeline-loop-unrolling$ ./run.sh
Time 0.163440, sum 100000000

Performance counter stats for 'taskset --cpu-list 0 ./prog':

    <not supported>          branch-misses:u

    0.390168664 seconds time elapsed

    0.261997000 seconds user
    0.120921000 seconds sys
```

```

Profiled target:  ./prog (PID 62559, part 1)
Events recorded:  Ir Bc Bcm Bi Bim
Events shown:     Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds:       99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation:  on

-----
Ir              Bc              Bcm              Bi              Bim
-----
2,000,139,295 (100.0%) 200,025,032 (100.0%) 2,944 (100.0%) 365 (100.0%) 174 (100.0%)  PROGRAM TOTALS
```

Vec_sum_v2()

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/pipeline-loop-unrolling$ ./run.sh
Time 0.183717, sum 100000000

Performance counter stats for 'taskset --cpu-list 0 ./prog':

    <not supported>          branch-misses:u

    0.428512065 seconds time elapsed

    0.300642000 seconds user
    0.120256000 seconds sys
```

```

Events shown:     Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds:       99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation:  on

-----
Ir              Bc              Bcm              Bi              Bim
-----
1,800,139,466 (100.0%) 125,025,075 (100.0%) 2,957 (100.0%) 365 (100.0%) 174 (100.0%)  PROGRAM TOTALS
```

Vec_sum_v3()

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/pipeline-loop-unrolling$ ./run.sh
Time 0.060451, sum 100000000
```

Performance counter stats for 'taskset --cpu-list 0 ./prog':

<not supported> branch-misses:u

0.295236731 seconds time elapsed

0.147374000 seconds user

0.137549000 seconds sys

```
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on
```

Ir	Bc	Bcm	Bi	Bim	
1,800,139,493 (100.0%)	125,025,081 (100.0%)	2,957 (100.0%)	365 (100.0%)	174 (100.0%)	PROGRAM TOTALS

Ускорение = $0.16/0.06 = 2.66$ раза (по сравнению с Vec_sum())

Ускорение = $0.18/0.06 = 3$ раза (по сравнению с Vec_sum_v2())

```
hope332@LostPoint1-2:/mnt/c/Users/79826/Desktop/paralel/lab1/pipeline-loop-unrolling$ ./run.sh
Time 0.034381, sum 100000000
```

Performance counter stats for 'taskset --cpu-list 0 ./prog':

<not supported> branch-misses:u

0.190789426 seconds time elapsed

0.057858000 seconds user

0.125359000 seconds sys

```
Events recorded: Ir Bc Bcm Bi Bim
Events shown: Ir Bc Bcm Bi Bim
Event sort order: Ir Bc Bcm Bi Bim
Thresholds: 99 0 0 0 0
Include dirs:
User annotated:
Auto-annotation: on
```

Ir	Bc	Bcm	Bi	Bim	
575,139,668 (100.0%)	125,025,110 (100.0%)	2,928 (100.0%)	366 (100.0%)	180 (100.0%)	PROGRAM TOTALS

Ускорение сохраняется при O2